Assignment on Server Configuration

SUBMITTED BY
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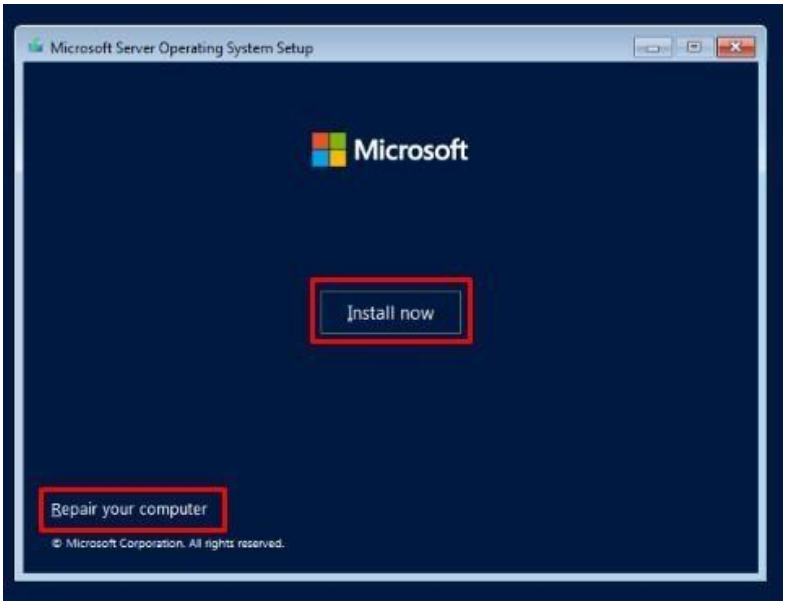
Assignments 1:

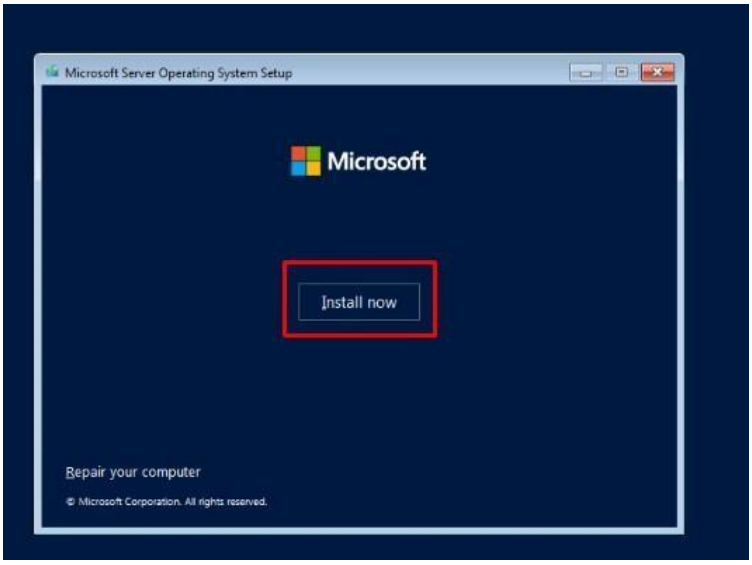
Requirment 1: Installation Troublshoot



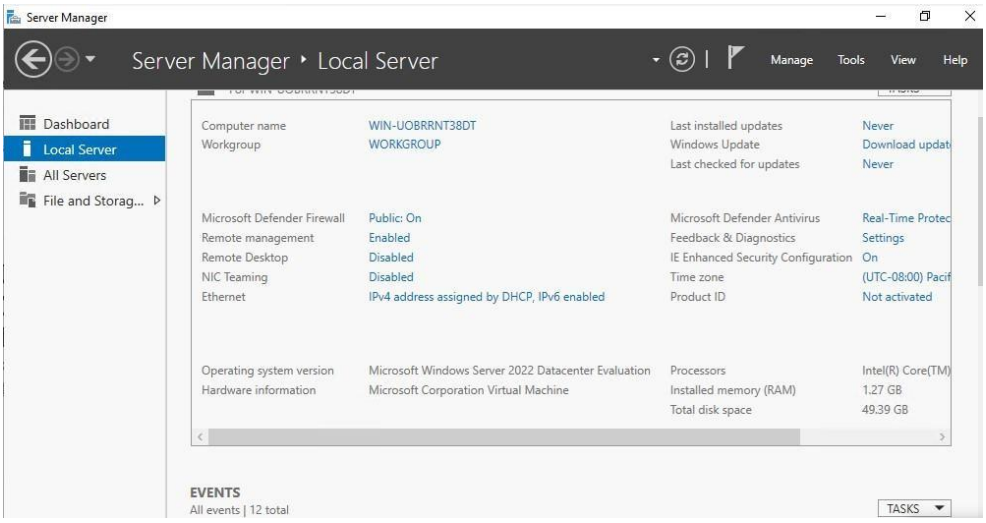
When **'Repair your computer'** is selected, usually enter a Recovery Environment, which comes with troubleshooting options. Namely:

- Startup Repair,
- System Restore,
- Uninstall Updates,
- System Image Recovery etc.

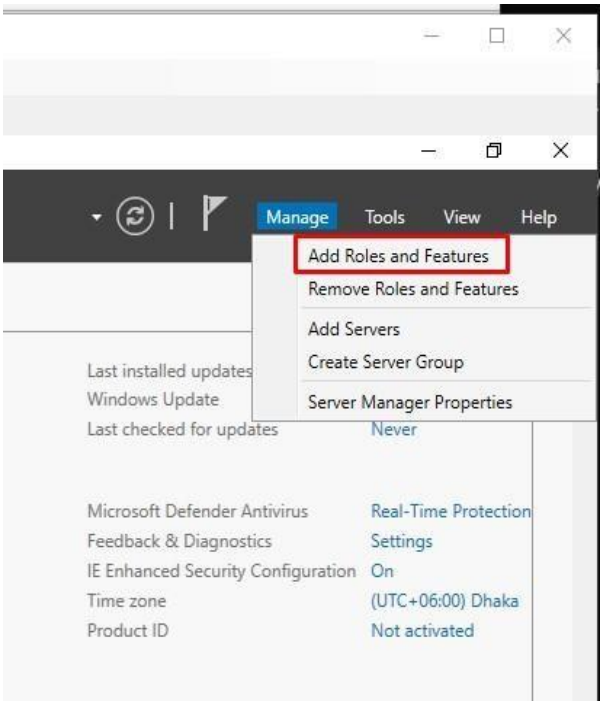
The main purpose of going to **'Repair your computer'** is to be able to resolve problems from outside the operating system, even if the server won't boot or has a serious problem. Since I want to set up a new operating system and there are no problems, I will click **Install Now.**



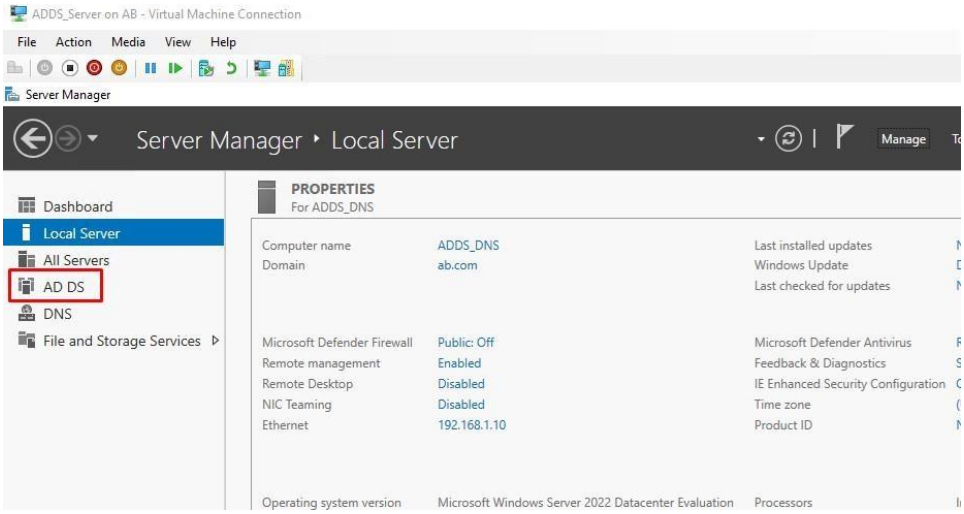
Now I have created a server by completing the next steps step by step.



Requirment 2: Configuring Various roles in Windows Server.
After setting up the server, need to decide what the server will do. These tasks are called roles. For example, a server can be a file server, or a web server, or act as a domain controller. Now let's **configure Active Directory Domain Services (ADDS)**. ADDS helps manage users, groups, and other resources on a network. The main function of the ADDS server is to **centrally manage all users**, computers, and other resources on a network. To configure ADDS, first go to Server Manager and install the ADDS role.



Now I have created AD DS server Roles Configure Succesfully.

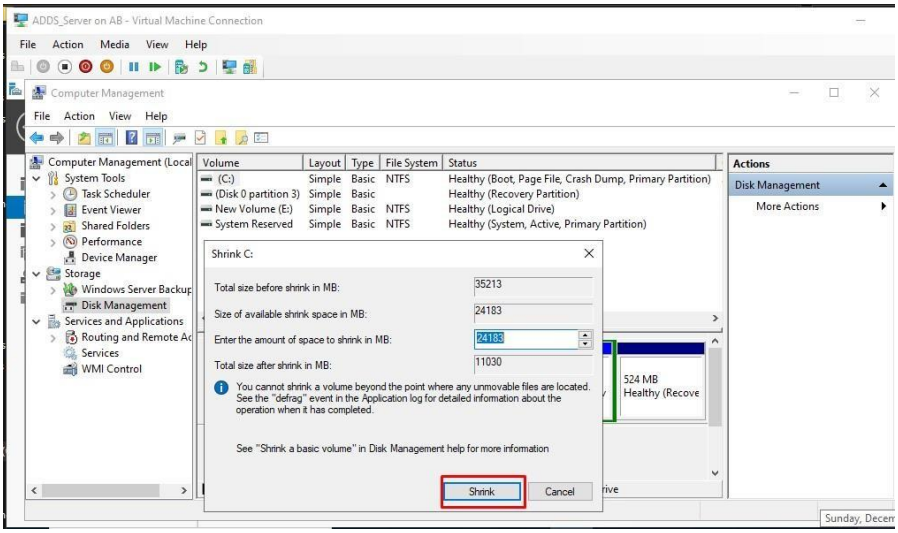


Assignment 2:

Working on Disk Management: 'Disk management' is the task of managing and partitioning a computer's disk drives. I want to **shrink** partitions, **extend** partitions, create **dynamic disks**, and create **RAID volumes** using Disk Management.

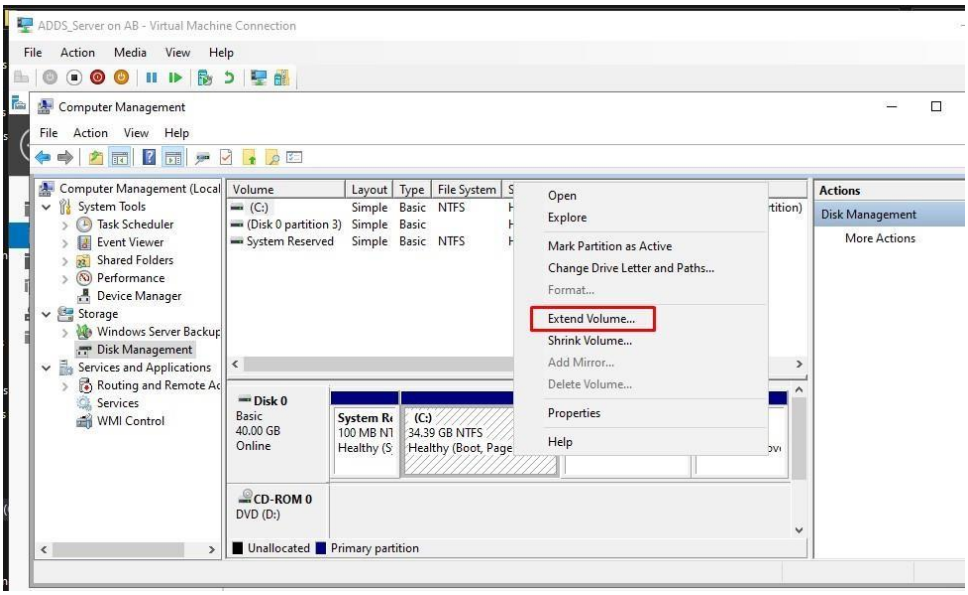
Shrinking partition Scenario

Objective: **Reducing the size** of an existing volume to make room for creating new partitions from unused space.
Mechanism: **Right-click** on the volume in the Disk Management console and select the Shrink Volume option.



Extend partition Scenario

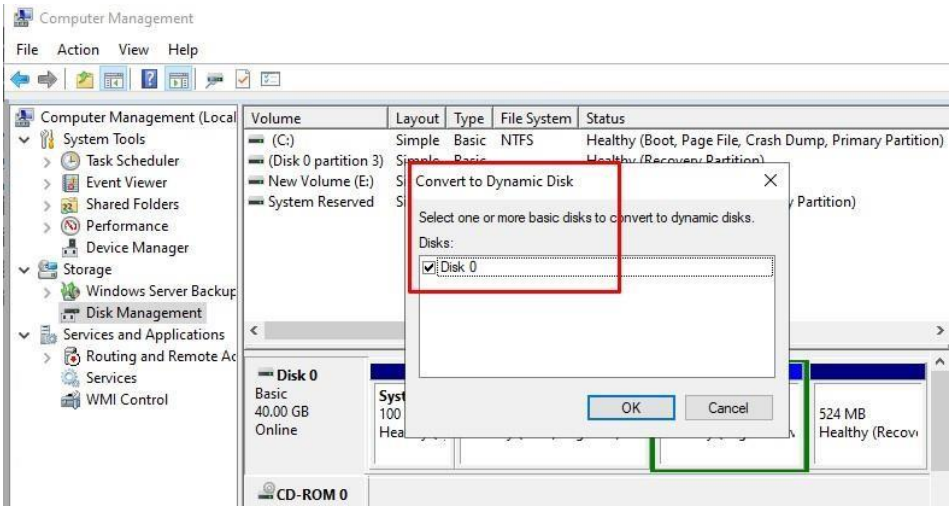
Objective: **Increasing** the size of a volume, which is added to the unused space (Unallocated Space) left behind that volume.
Mechanism: Right-click on the volume in the Disk Management console and use the Extend Volume option.



Working on Dynamic Disk

Objective: **Increase storage flexibility** by making volumes spanned, mirrored, or striped.

Mechanism: First, open the Disk Management tool. There, right-click on the disk I want to convert to dynamic and select the "Convert to Dynamic Disk" option. Then follow the instructions in the wizard to convert the disk to dynamic. Converting a basic disk to a dynamic disk in the Disk Management console.



Working on RAID Voluam Types

RAID is a method of using redundant disks, which can increase data security and performance. The main RAID volume types are:

RAID 0 (striping): This increases data access speed by striping data across multiple disks,but does not provide data security.

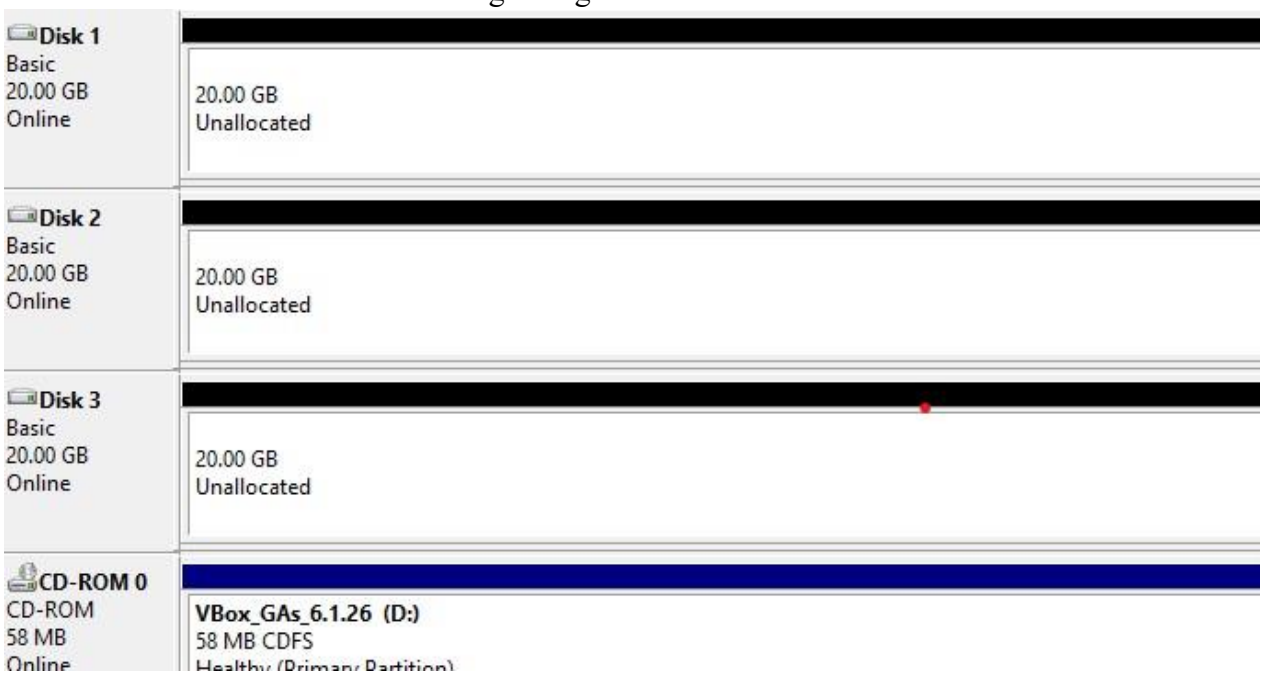
RAID 1 (Mirroring): It increases data security by mirroring data across two disks, but half of the disk capacity is lost.

RAID 5: It increases data security and performance by stripe data and parity information across multiple disks.

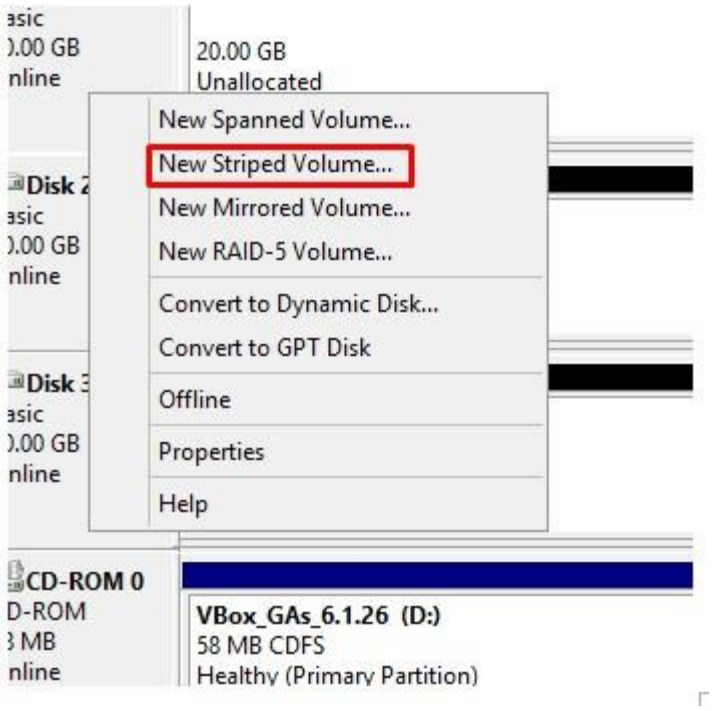
RAID 10: This is a combination of RAID 1 and RAID 0, which provides higher levels of data security and performance.

Mechanism: (Performance) Increase. Create RAID 0 (Striped), RAID 1 (Mirrored), or RAID 5 (with Parity) volumes by combining multiple disks using Disk Management. Each red volume configuration has specific steps. To configure a RED volume, first need to open the Disk Management tool. Then, need to convert the extra disks on the system to dynamic disks. Once configured, those disks can be used to create different RED volumes, such as striped, mirrored, or other RED volumes, including Parrot.

Step 1: Open disk management(**diskmgmt.msc**). It opens the Disk Management console. three Virtual Hard disks attach to the existing storage.



Step 2: Select one of the disks, right-click on it, and **select New Striped Volume.** The Stripped Volume is for RAID 0 configuration**.** A new simple volume wizard would open to configure its settings.



Step 3: Add two disks for **RAID 0 configuration**. adding to the disk because at least require two disks for the RAID 0 configuration.

Step 4: Type the amount of **space** as like **2048 MB >2 GB** and click on **Next**. The space that would be used on each disk to make the data mirrored from one disk to the other.

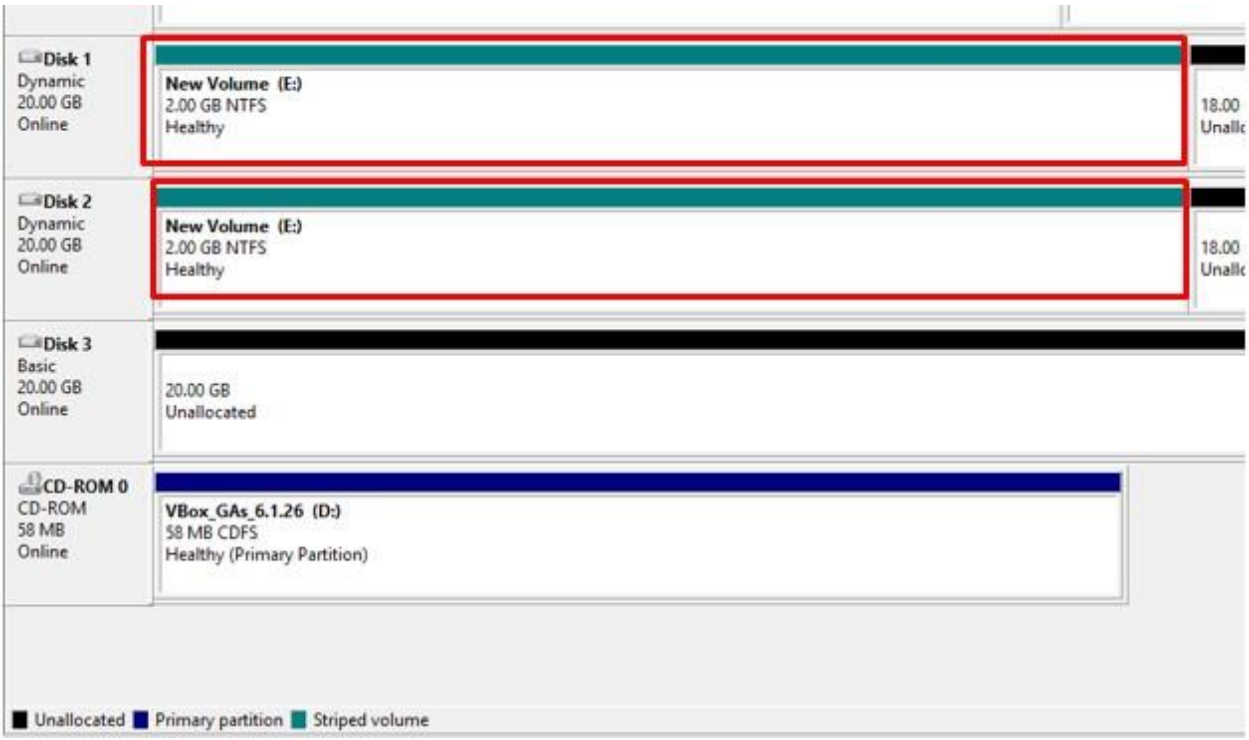
Step 5: Assign a **drive letter** and click **Next**. It provides the name of the partition so that, it could be distinguished.

Step 6: Select the partitioning as **NTFS** and click **Next.** We'll go with the default option in the scenario, and if required we can change it.

Step 7: Confirmation wizard would appear to click on **Finish**. It contains all the settings list that you applied for the configuration.

Step 8: Click on **Yes to complete the creation of RAID 0 volume.** It can take some time to make the changes.

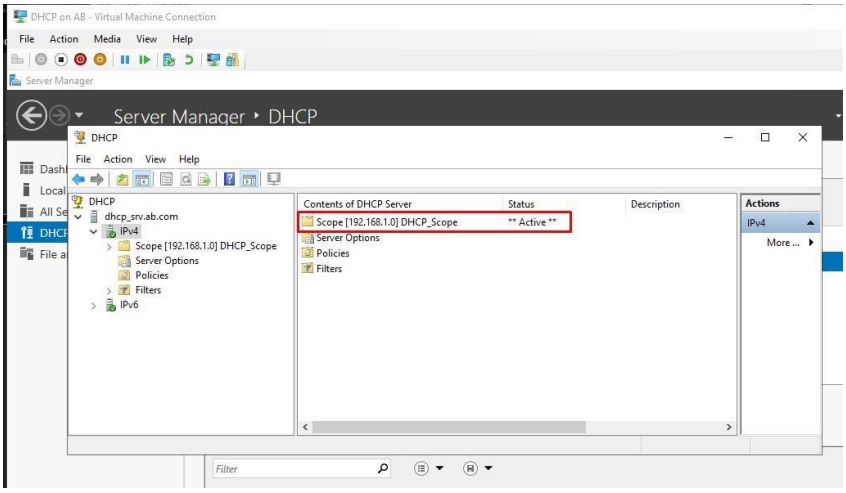
RAID 0 Volume of 2 GB is created with Disk 1 and Disk 2. This means whenever data is stored on a Disk 1 then the same could be seen in Disk 2 as well.



Assignment 3:

DHCP Concept: DHCP is a protocol that automatically assigns IP addresses to users. Simply put, it allows the Internet to be used for network busses.
Objective: Automatically assign IP addresses, subnet masks, and gateways to devices on the network.
Mechanism: The DHCP server assigns IP addresses to clients using the IP Lease process (Discover, Offer, Request, Acknowledge).

Scope Options:
Objective: Creating a specific IP range and Defining a pool of IP addresses on a specific network subnet.
Mechanism: Set the IP range and subnet mask using the New Scope wizard under IPv4 or IPv6 in the DHCP Console. Right-click on IPv4, follow the New Scope wizard, and set the start/end IP addresses.

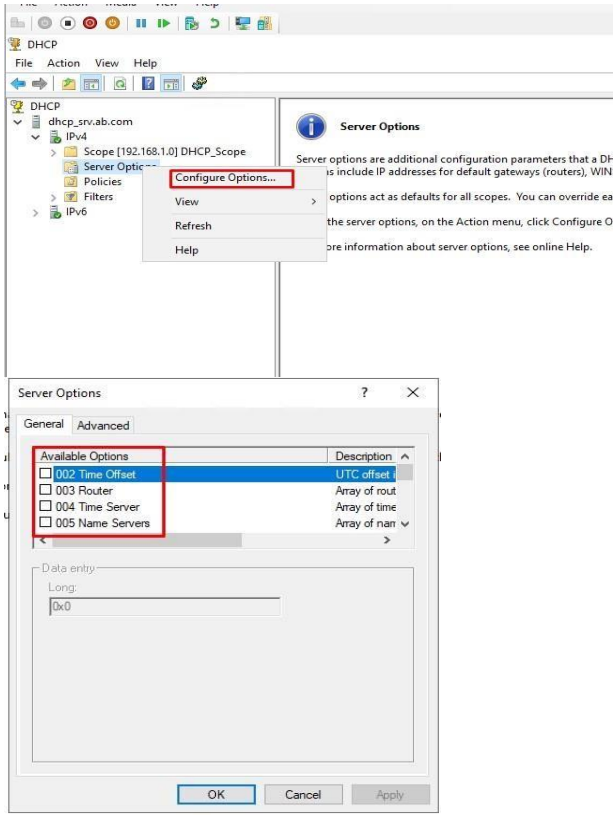


DHCP options:
DHCP options basically give client devices the settings they need to function properly on the network.
Objective: Providing the client with additional configuration along with an IP address.
Mechanism: Right-click on Scope Options under Scope and select Configure Options. Here, options like Router (003) (default gateway) and DNS Servers (006) are set.

003 Router (Default Gateway): Provides the IP address of the router to the client for sending traffic outside the network.

006 DNS Servers: Helps convert domain names to IP addresses.

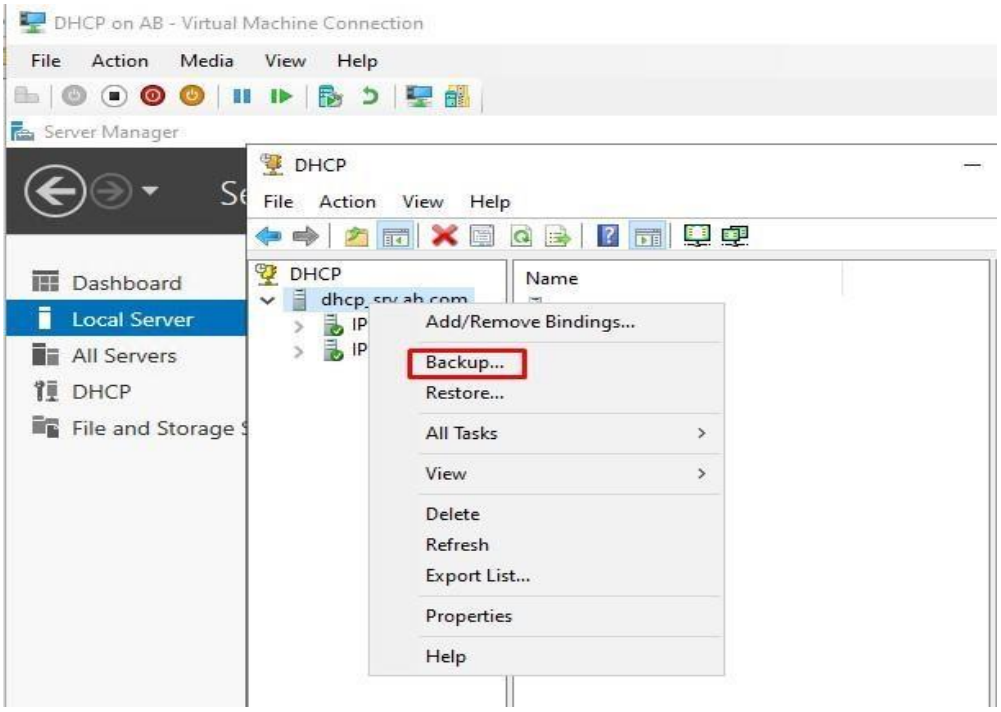
015 DNS Domain Name: Sets the domain name for clients.

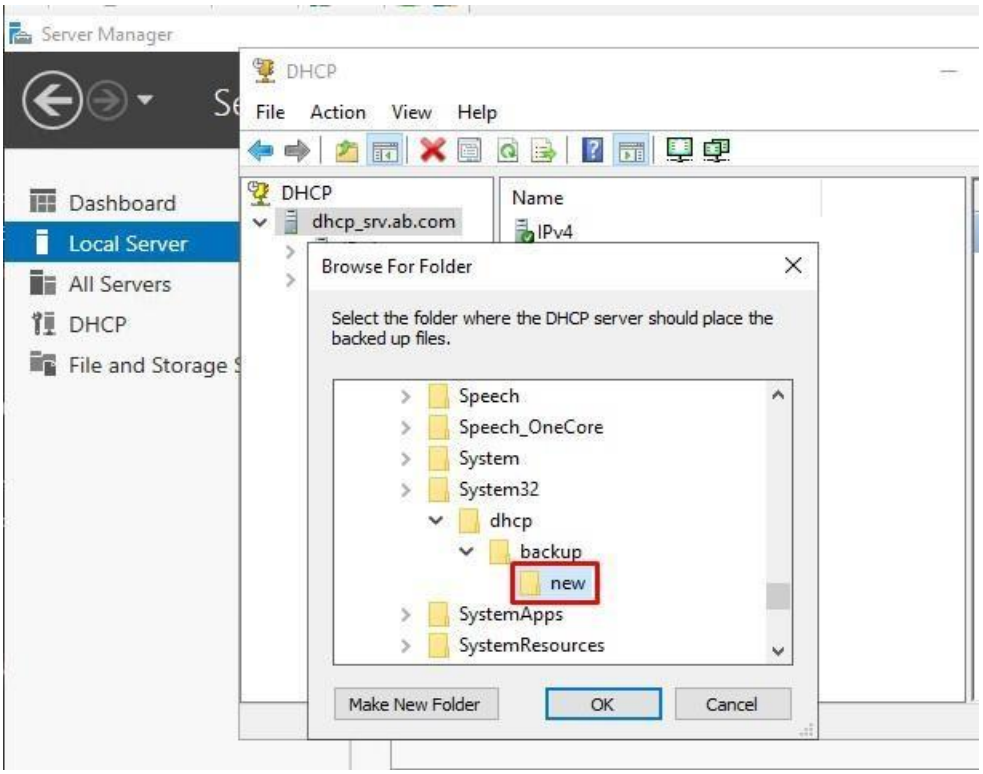


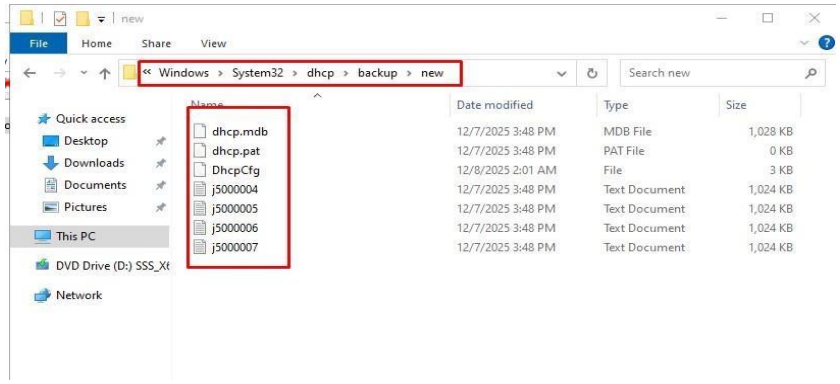
DHCP backup

Objective: **Protecting** DHCP configuration data and lease records.

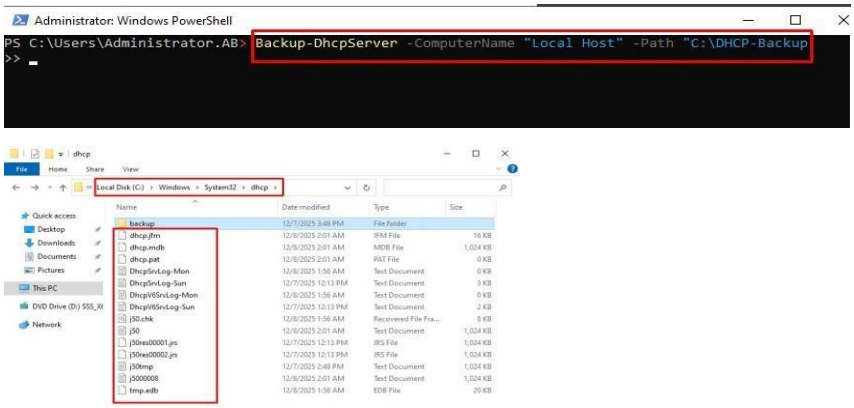
Mechanism: Right-click on the server node in the DHCP Console and select the Backup option.







Backup in PowerSell:

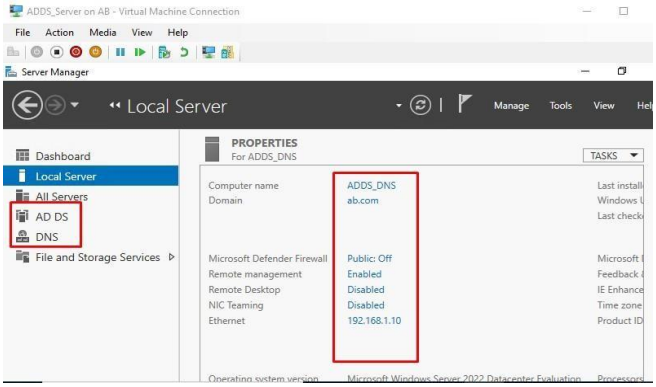


Assignment 4: Active Directory

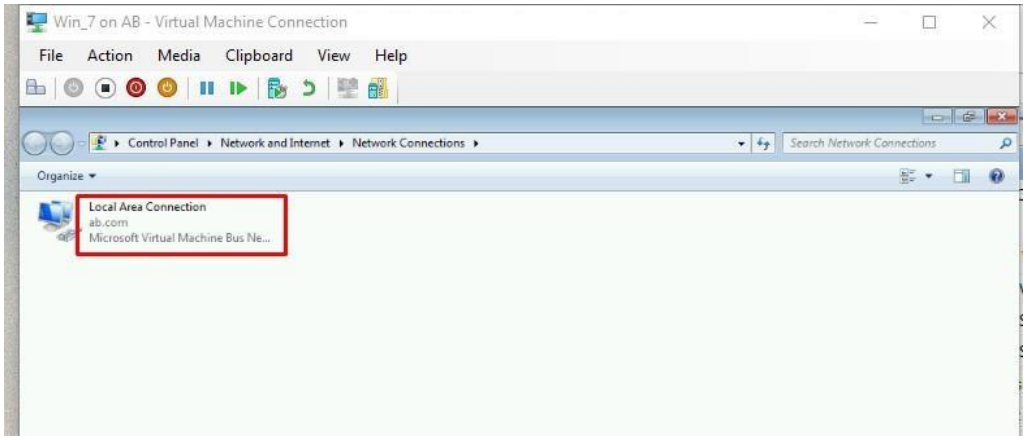
Centralize Administration and Working with AD. Centralized Administration (CA) means controlling all users, computers, and security policies on a network from a single location. Active Directory is the main tool for implementing this CA.

Objective: Bringing a computer under the domain's central authentication framework.

Mechanism: Creating a forest and domain by installing the Active Directory Domain Services (AD DS) role and creating user/computer objects.

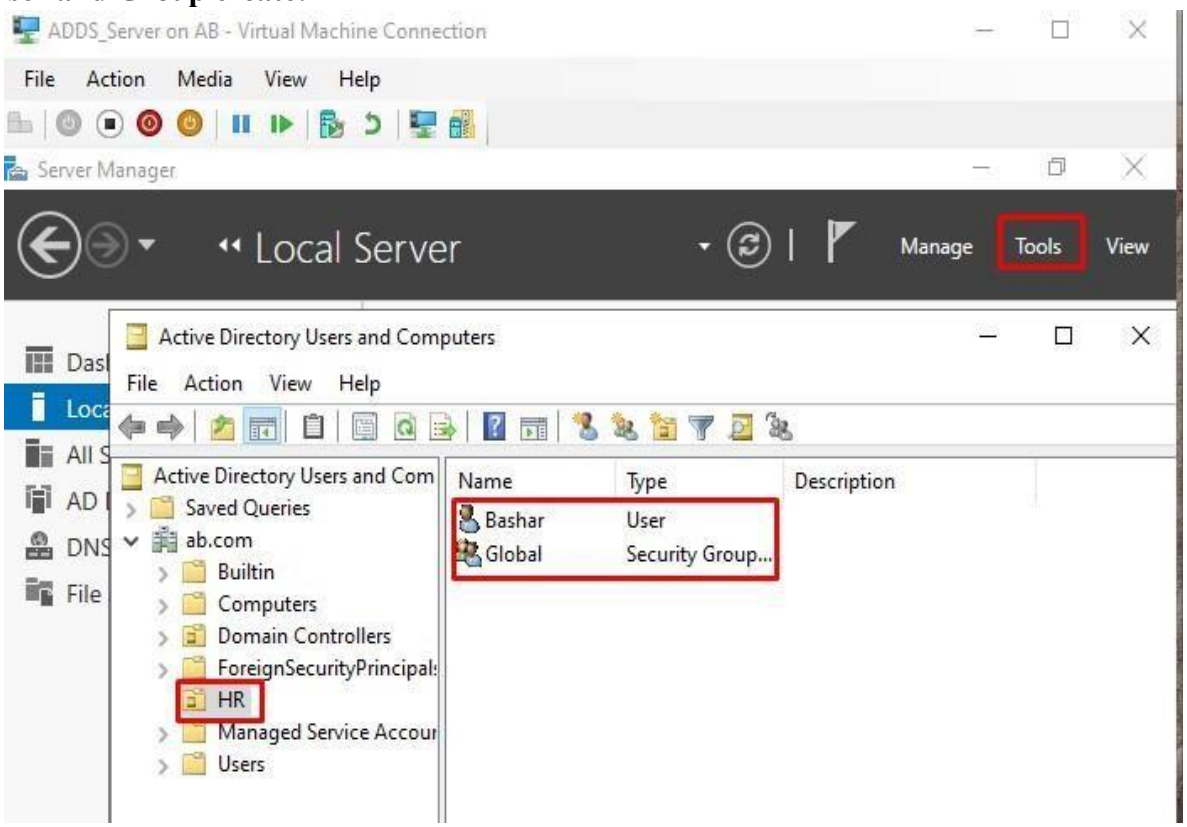


adding users -



Objective: Ensuring central management and authentication of network users, computers, and other resources.
Mechanism: Using the AD DS forest, domain, and domain controller (DC) structure. The most important tool in AD, used for user, group, and OU management.

User and Group create:



A set of rules that you can apply to computers or users. An OU is a part of Active Directory where you can organize your server computers or users. **View list:**

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADUser -Filter * -Properties Name, SamAccountName, Enabled


DistinguishedName : CN=Administrator,CN=Users,DC=ab,DC=com
Enabled           : True
GivenName         :
Name              : Administrator
ObjectClass       : user
ObjectGUID        : 2527e582-30f7-4878-9e15-b0a82252adaa
SamAccountName    : Administrator
SID               : S-1-5-21-2978459849-1215766918-1402864147-500
Surname           :
UserPrincipalName :

DistinguishedName : CN=Guest,CN=Users,DC=ab,DC=com
Enabled           : False
GivenName         :
Name              : Guest
ObjectClass       : user
ObjectGUID        : 4892a0c7-c85d-48fc-8ff2-cb2bace4588b
SamAccountName    : Guest
SID               : S-1-5-21-2978459849-1215766918-1402864147-501
Surname           :
UserPrincipalName :

DistinguishedName : CN=krbtgt,CN=Users,DC=ab,DC=com
Enabled           : False
```

This entire setup here shows implementing Centralized Administration using Active Directory (AD DS) and Networking Services (DHCP, DNS) in Windows Server 2022.

Why is GPO used?

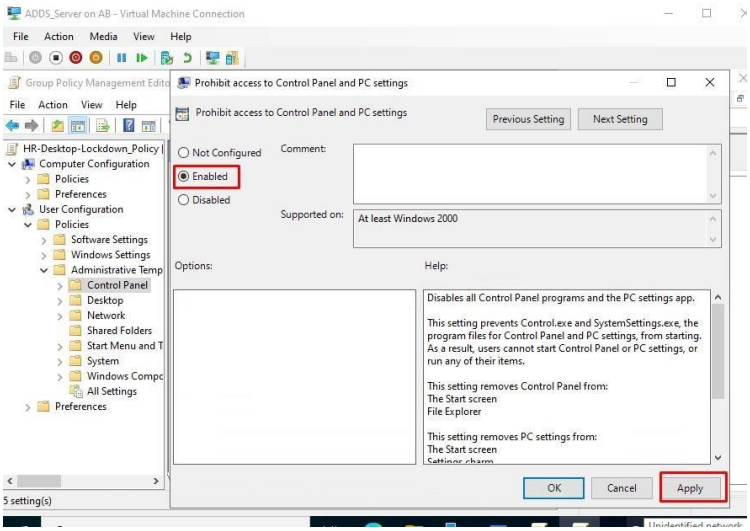
Objective: Automatically and consistently apply security policies and configuration settings to all users and computers.

Mechanism: GPOs manage security, user interface, software installation, and power settings.

Group Policy (GPO) is used to enforce rules and policies on computers and users on a network. This allows to configure various settings from a central location, which saves time and increases security. In short, GPO makes network management easier and more secure.

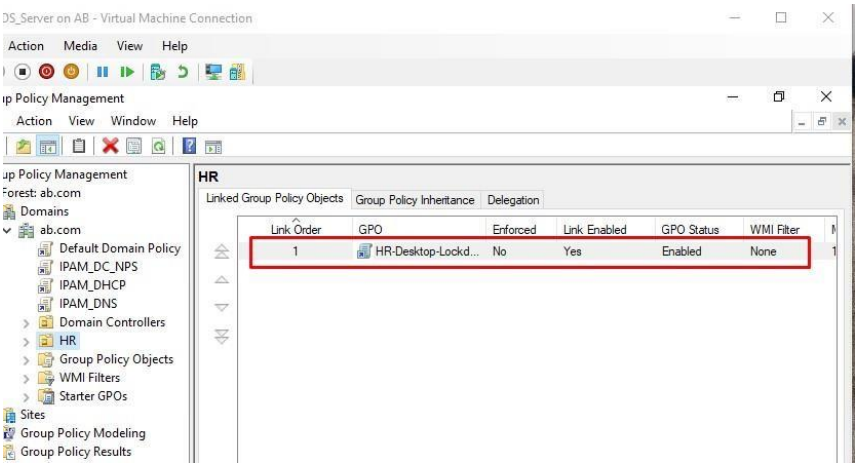
GPO implementation on OU

Objective: Applying policies to a subset of specific users or computers in the domain.
Mechanism**:** **Create GPO** using the Group Policy Management Console and **link** it to an Organizational Unit (OU).
GPO-1:



The Prohibit access to Control Panel and PC settings policy is Enabled under User Configuration > Policies > Administrative Templates > Control Panel. This will prevent users in the HR OU from accessing Control Panel and PC Settings from their computers. This will prevent users from changing unnecessary or unauthorized settings.

GPO-2: To enforce security and/or operational restrictions (desktop lockdown settings) on the computers and user accounts that belong to the Human Resources (HR) department.



All rules defined in the HR-Desktop-Lockd GPO (possibly related to desktop lockdown) are currently applied to all users and computers in the HR organizational unit. A GPO named HRDesktop-Lockd is linked to this HR OU (Link Enabled: Yes).
Impact: Desktop Lockdown Settings prevent HR department users from performing any tasks on their desktop or changing any settings. Will restrict running certain applications.

OVERVIEW

1. Installation and Troubleshooting

- Server Setup: The process starts by selecting 'Install now' on the Microsoft Server Operating System Setup screen to set up the new operating system.
- Troubleshooting: The 'Repair your computer' option is used to enter a Recovery Environment, offering options like Startup Repair and System Restore, primarily to resolve problems from outside the operating system, even if the server won't boot.

2. Configuring Various Server Roles

- **Active Directory Domain Services (ADDS):**
  - Purpose: ADDS helps centrally manage all users, computers, and other resources on a network.
  - Method: The ADDS role is installed via the Server Manager by selecting 'Add Roles and Features' under the 'Manage' menu.
- **Disk Management:**
  - Objective: To manage and partition a computer's disk drives.
  - The document demonstrates:
  - Shrinking Partition: Reducing the size of a volume to create unused space for new partitions.
  - Converting to Dynamic Disk: Increasing storage flexibility by converting a basic disk to a dynamic disk.
  - Creating RAID Volumes: Using multiple disks to increase data security and performance (e.g., RAID 0 Striped Volume is created using Disk 1 and Disk 2 for performance).

3. DHCP Concept and Configuration

- **DHCP Server:**

Concept: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices on the network.

- Scope Options: Additional configuration parameters like the Router (Default Gateway) (003) and DNS Servers (006) are provided to client devices along with the IP address.
- DHCP Backup: The configuration data and lease records are protected by performing a backup through the DHCP Console.

4. Active Directory and GPO

- Centralized Administration (CA): Active Directory is the primary tool for implementing CA, which means controlling all users, computers, and security policies from a single location.
- User and Group Management: The AD DS forest and domain structure are used to ensure the central management and authentication of network users.
- Group Policy Object (GPO) Usage:

Purpose: GPO is used to enforce rules and policies on computers and users on a network to apply security policies and configuration settings automatically and consistently.

- GPO-1 (Control Panel Restriction): A policy to Prohibit access to Control Panel and PC settings is Enabled under User Configuration to prevent users in the HR OU from changing unauthorized settings.

- GPO-2 (Desktop Lockdown): The HR-Desktop-Lockd GPO is linked to the HR OU to enforce security and/or operational restrictions (desktop lockdown settings), which will restrict HR users from performing certain tasks or changing settings.

Thank You